

MACHRIHANISH CAMPBELTOWN, United Kingdom

WMO: 031110

Lat: 55.4408N

Lon: 5.6969W

Elev: 10

StdP: 101.20

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.6	-0.9	-5.6	2.4	-0.3	-4.2	2.7	1.0	19.2	6.2	17.7	6.0	2.2	100	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.7	20.8	16.4	19.0	15.6	17.8	14.9	17.2	19.5	16.3	18.2	15.6	17.1	5.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.6	18.1	15.5	11.0	17.2	14.8	10.5	16.4	48.4	19.5	45.9	18.4	43.6	17.2	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	14.3	12.8	DB	-5.3	23.9	2.0	2.2	-6.8	25.5	-8.0	26.8	-9.2	28.0	-10.6	29.6	(4)
(5)			WB	-5.9	19.0	2.0	1.2	-7.4	19.9	-8.5	20.6	-9.7	21.3	-11.1	22.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	5.5	5.5	6.4	7.9	10.3	12.6	14.2	14.4	13.0	10.6	7.8	5.9	(6)
(7)		DBStd	4.07	2.60	2.66	2.40	2.35	2.40	2.04	1.87	1.79	1.94	2.44	2.79	3.23	(7)
(8)		HDD10.0	703	139	128	114	71	24	2	0	0	2	23	73	129	(8)
(9)		HDD18.3	3216	397	361	371	313	250	172	129	124	159	241	315	385	(9)
(10)		CDD10.0	533	1	1	2	7	33	80	131	135	93	40	8	2	(10)
(11)		CDD18.3	3	0	0	0	0	0	1	2	1	0	0	0	0	(11)
(12)		CDH23.3	7	0	0	0	0	1	1	4	2	0	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	6.2	7.6	7.3	6.7	6.1	5.6	5.0	4.7	5.0	5.8	6.7	7.0	7.1	(14)
(15)	Precipitation	PrecAvg	1307	138	111	97	83	76	78	85	98	99	149	149	143	(15)
(16)		PrecMax	1568	253	214	182	150	142	154	146	219	194	286	236	282	(16)
(17)		PrecMin	1003	20	36	38	39	16	21	22	34	19	41	81	58	(17)
(18)		PrecStd	161	51	51	34	33	36	33	29	46	42	66	45	55	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	12.7	13.9	17.2	21.0	22.0	23.5	22.7	19.8	16.7	14.7	13.2	(19)
(20)			MCWB	11.0	11.3	11.1	12.2	15.6	16.0	18.1	17.8	16.7	14.5	13.3	11.7	(20)
(21)		2%	DB	11.1	11.1	12.0	14.4	17.7	19.2	20.7	19.9	18.0	15.5	13.4	12.2	(21)
(22)			MCWB	10.0	9.7	9.8	10.9	13.1	15.1	16.6	16.6	15.4	13.9	12.2	11.0	(22)
(23)		5%	DB	10.1	10.1	10.7	12.6	15.7	17.3	18.8	18.3	16.8	14.7	12.5	11.2	(23)
(24)			MCWB	9.1	8.9	8.9	9.9	12.1	14.2	15.6	15.6	14.6	13.2	11.3	10.1	(24)
(25)		10%	DB	9.3	9.3	9.8	11.3	14.0	15.9	17.3	17.2	16.0	14.0	11.8	10.3	(25)
(26)			MCWB	8.3	8.1	8.4	9.2	11.2	13.4	14.9	14.9	14.0	12.6	10.6	9.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	11.6	11.5	13.1	15.9	17.1	18.9	18.6	17.4	15.2	13.7	12.1	(27)
(28)			MCD B	12.2	12.4	13.0	15.9	20.3	20.6	22.6	21.4	18.9	16.1	14.3	12.8	(28)
(29)		2%	WB	10.3	10.1	10.3	11.3	13.9	15.7	17.2	17.2	16.1	14.3	12.5	11.2	(29)
(30)			MCD B	11.0	10.7	11.5	13.6	16.7	18.2	19.6	19.1	17.3	15.2	13.2	12.0	(30)
(31)		5%	WB	9.4	9.2	9.4	10.3	12.7	14.7	16.2	16.3	15.2	13.6	11.6	10.3	(31)
(32)			MCD B	10.1	9.9	10.4	12.1	15.0	16.7	18.2	17.7	16.2	14.5	12.4	11.2	(32)
(33)		10%	WB	8.5	8.4	8.6	9.5	11.7	13.8	15.3	15.5	14.5	12.9	10.7	9.4	(33)
(34)			MCD B	9.3	9.2	9.6	11.0	13.5	15.5	16.9	16.7	15.5	13.8	11.6	10.2	(34)
(35)	Mean Daily Temperature Range		MDBR	4.4	4.8	5.6	6.1	6.5	5.9	5.7	5.5	5.1	4.7	4.8	(35)	
(36)		5% DB	MCD BR	4.6	4.6	6.1	7.7	9.0	8.1	8.3	7.0	6.1	5.1	4.6	4.9	(36)
(37)			MCWBR	4.6	4.1	4.7	5.0	5.5	5.0	5.1	4.5	4.3	4.3	4.4	4.8	(37)
(38)			5% WB	MCD BR	4.5	4.5	5.5	6.7	7.8	7.2	7.1	6.2	5.4	4.6	4.5	4.8
(39)		MCWBR		4.6	4.3	4.6	4.7	5.2	4.8	4.8	4.4	4.3	4.3	4.6	4.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.338	0.349	0.370	0.384	0.387	0.380	0.384	0.368	0.365	0.348	0.330	0.326	(40)	
(41)		taud	2.230	2.323	2.320	2.325	2.368	2.421	2.417	2.473	2.476	2.440	2.306	2.197	(41)	
(42)		Ebn at Noon	556	715	792	837	856	865	854	847	798	714	570	471	(42)	
(43)		Edh at Noon	59	77	97	111	112	108	106	95	83	68	54	49	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.50	1.18	2.31	3.77	4.83	4.79	4.56	3.70	2.56	1.39	0.66	0.37	(44)	
(45)		RadStd	0.06	0.13	0.18	0.28	0.44	0.50	0.42	0.27	0.18	0.17	0.07	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.25	N/A	+1.00	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page